3.2.8 AC Series—Parallel Circuits

Consider the series–parallel circuit consisting of three branches A, B, and C as shown. In Fig. 3.59.

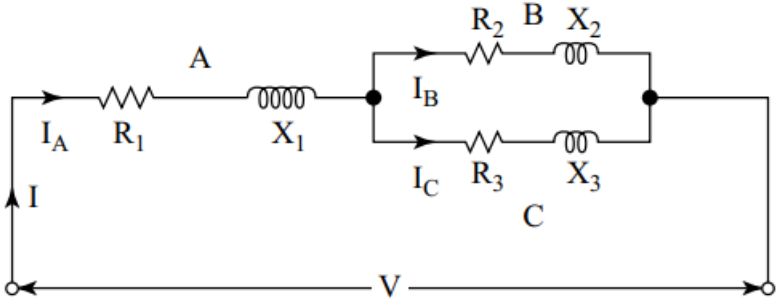


Figure 3.59

Impedance of branch A, $Z_A = R_1 + jX_1$

Impedance of branch B, $Z_B = R_2 + jX_2$

Impedance of branch C, $Z_C = R_3 + jX_3$

Total impedance of the circuit Z is

$$Z = Z_A + \frac{Z_B Z_C}{Z_B + Z_C}$$

Total current, $$I = \frac{V}{Z} = I_A$$

Current, $$I_B = I \frac{Z_C}{Z_B + Z_C}$$ (applying current divider rule)

Current, $$I_C = I \frac{Z_B}{Z_B + Z_C}$$

By applying the admittance method we can also solve the problem as

$$Y_A = \frac{1}{Z_A}; Y_B = \frac{1}{Z_B}; Y_C = \frac{1}{Z_C}$$

Total admittance of the parallel branches B and C

$$Y_{BC} = Y_B + Y_C$$

Impedance,

$$Z_{BC} = \frac{1}{Y_{BC}}$$

Total impedance, $Z = Z_A + Z_{BC}$

Total current, $I = \frac{V}{Z} = \frac{V}{Z_A + Z_{BC}}$

Example 3.28 Determine the total current drawn from the supply by the series–parallel circuit shown in Fig. 3.60. Also calculate the power factor of the circuit.

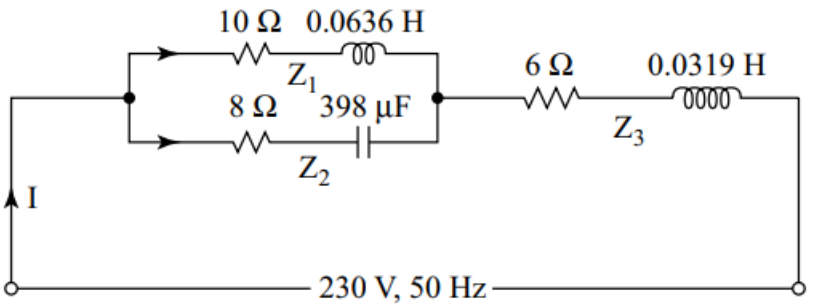


Figure 3.60

$$\omega = 2\pi f = 2 \times 3.14 \times 50 = 314$$

$$Z_1 = 10 + j314 \times 0.0638 = 10 + j20 = 22.36 \angle 64°$$

$$Z_2 = 8 - jX_C$$

$$X_C = \frac{1}{\omega C} = \frac{10^6}{314 \times 398} = 8\,\Omega$$

$$Z_2 = 8 - j8 = 11.3 \angle -45°$$

$$Z_3 = 6 + j314 \times 0.0319 = 6 + j10 = 11.66 \angle 59°$$

Equivalent impedance, $$Z = \frac{Z_1 Z_2}{Z_1 + Z_2} + Z_3$$

$$= \frac{22.36 \angle 64° \times 11.3 \angle -45°}{10 + j20 + 8 - j8} + 11.66 \angle 59°$$

$$= \frac{252.7 \angle 19°}{18 + j12} + 11.66 \angle 59°$$

$$= \frac{252.7 \angle 19°}{21.63 \angle 34°} + 11.66 \angle 59°$$

$$= 11.68 \angle -15° + 11.66 \angle 59°$$

$$= 11.21 - j3 + 6 + j10$$

$$= 17.21 + j7$$

$$= 18.58 \angle 22° \, \Omega$$

Current, $$I = \frac{V}{Z} = \frac{230 \angle 0°}{18.58 \angle 22°} = 12.37 \angle -22° \text{ A}$$

Total current, $I = 12.37$ A

Power factor, $\cos \phi = \cos 22° = 0.92$ lagging

Example 3.30 In the series–parallel circuit shown in Fig. 3.62, the parallel branches A and B are in series with branch C. The impedances are $Z_A = (4 + j3)\ \Omega$, $Z_B = (10 - j7)\ \Omega$ and $Z_C = (6 + j5)\ \Omega$. If the voltage applied to the circuit is 200 V at 50 Hz, calculate (a) current I_A, I_B, and I_C; (b) the power factor for the whole circuit. Draw also the phasor diagram.

Solution:

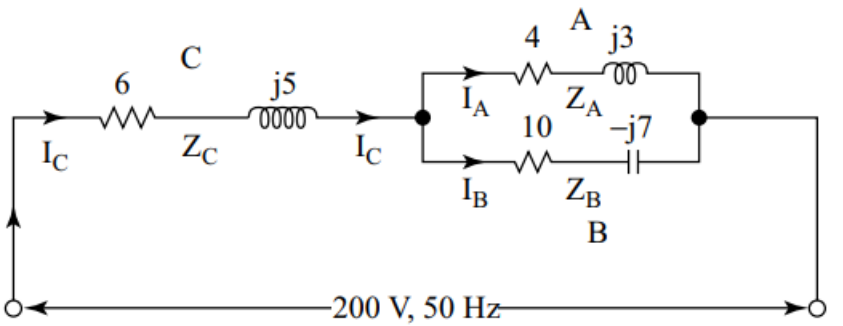


Figure 3.62

$$Z_A = (4 + j3) = 5\angle 36.9°\ \Omega$$

$$Z_B = (10 - j7) = 12.2\angle -35°\ \Omega$$

$$Z_C = (6 + j5) = 7.8\angle 39.8°\ \Omega$$

$$Z_A + Z_B = 4 + j3 + 10 - j7$$

$$= 14 - j4$$

$$= 14.56\angle -16°$$

$$Z_{AB} = \frac{Z_A Z_B}{Z_A + Z_B}$$

$$= \frac{5\angle 36.9° \times 12.2\angle -35°}{4 + j3 + 10 - j7}$$

$$= \frac{5\angle 36.9° \times 12.2\angle -35°}{14 - j4} = \frac{61\angle 1.9°}{14.56\angle -16°}$$

$$= 4.19\angle 17.9°$$

$$= 4.19\,(\cos 17.9° + j\sin 17.9°)$$

$$= 4 + j1.3$$

$$Z = Z_C + Z_{AB} = 6 + j5 + 4 + j1.3$$

$$= 10 + j6.3 = 11.8\angle 32.2°$$

Let $V = 200\angle 0°$

$$I_C = \frac{V}{Z} = \frac{200\angle 0°}{11.8\angle 32.2°} = 16.35\angle -32.2° \text{ A}$$

Using current divider rule,

$$I_A = I_C \frac{Z_B}{Z_A + Z_B} = 16.35\angle -32.2° \times \frac{12.2\angle -35°}{14.56\angle -16°} = 13.7\angle -51.2\,A$$

$$I_B = I_C \frac{Z_A}{Z_A + Z_B} = 16.35\angle -32.2° \times \frac{5\angle 36.9°}{14.56\angle -16°} = 5.7\angle 20.7°\,A$$

Phase angle between applied voltage V and line current I is $-\ 32.2°$

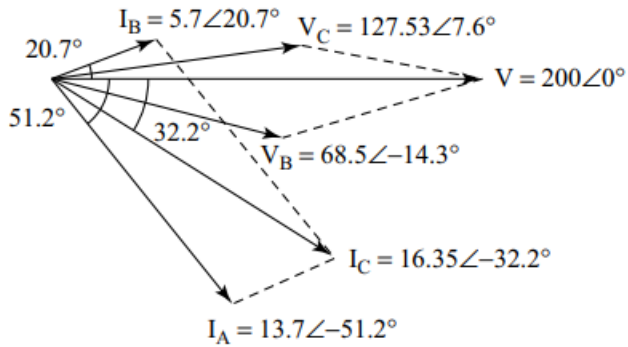
Hence, power factor of the whole circuit = $\cos\phi = \cos 32.2°$
$= 0.846$ lagging

Voltage drop across series branch C, $V_C = I_C Z_C$
$= 16.3\angle -32.2° \times 7.8\angle 39.8°$
$= 127.53\angle 7.6°\,V$

Voltage drop across parallel branches, $V_A = V_B = I_C \cdot Z_{AB}$
$= 16.35\angle -32.2° \times 4.19\angle 17.9$
$= 68.5\angle -14.3°\,V$

Note that voltage across the parallel branches is also equal to $I_A Z_A$ or $I_B Z_B$.

The complete phasor diagram is shown in Fig. 3.63.



Example 3.31 In the circuit shown in Fig. 3.64, determine the voltage at 50 Hz to be applied across terminals AB in order that a current of 10 A flows in the capacitor.

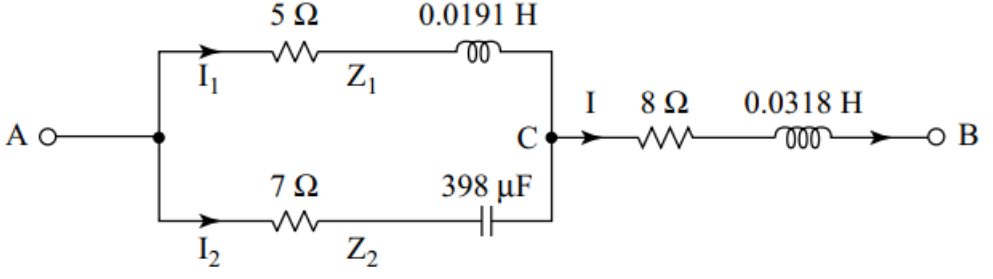


Figure 3.64

Solution:

$$Z_1 = 5 + j2\pi \times 50 \times 0.0191 = 5 + j6 = 7.81\angle 50.2^\circ\ \Omega$$

$$Z_2 = 7 - j\frac{1}{2\pi \times 50 \times 398 \times 10^{-6}} = (7 - j8)\,\Omega = 10.63\angle -48.8^\circ\ \Omega$$

$$Z_3 = 8 + j2\pi \times 50 \times 0.0318 = (8 + j10)\,\Omega = 12.8\angle 51.34^\circ\ \Omega$$

Current in the capacitive branch,

$$I_2 = 10\angle 0^\circ = 10 + j0\ \text{A}$$

Voltage drop across the parallel branch

$$\begin{aligned} V_{AC} &= I_2 Z_2 \\ &= 10\angle 0^\circ \times 10.63\angle -48.8^\circ \\ &= 106.3\angle -48.8^\circ\ \text{V} \\ &= (70.02 - j79.98)\,\text{V} \end{aligned}$$

Current in inductive branch

$$I_1 = \frac{V_{AC}}{Z_1} = \frac{106.3\angle -48.8^\circ}{7.81\angle 50.2^\circ} = 13.6\angle -99^\circ\ \text{A}$$

$$= (-2.13 - j13.44)\,\text{A}$$

Circuit current, $I = I_1 + I_2 = 10 + j0 - 2.13 - j13.44 = 7.87 - j13.44$

$= 15.57 \angle -59.65° \text{ A}$

Voltage drop across series branch, $V_{CB} = IZ_3$

$= 15.57 \angle -59.65° \times 12.8 \angle 51.34°$

$= 199.4 \angle -8.31° \text{ V}$

Voltage applied across terminals AB, $V_{AB} = V_{AC} + V_{CB}$

$= (267.33 - j108.8) \text{ V}$

$= 288.62 \angle 22.15° \text{ V}$